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RESEARCH ARTICLE



Sociology of innovation: past and prospects

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ABSTRACT

The sociology of innovation is considered as a component of various research subfields within the social sciences, primarily falling under the umbrella of science and technology studies (STS). This research aims to explore the possibility, in the current era of global societal challenges caused by technologically and socially disruptive innovations, of reconceptualizing the sociology of innovation as an independent research subdiscipline within sociology, becoming an essential factor for social betterment. Within this framework, the research initially delineates how innovation has evolved into one of the key determinants of society. Drawing upon the integrative literature approach, this study subsequently explores the theoretical foundations of the sociology of innovation and its historical embeddedness in neighbouring disciplines, up to the current theories of socio-technical transformation. Drawing on Burawoy's concepts of public sociology serving the interests of civil society, the remaining sections highlight future perspectives of the sociology of innovation, attempting to establish analytical distinctiveness of its research subject and functions from other subdisciplines.

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1. Introduction

Since the advent of industrial society driven by scientific and technological breakthroughs, innovation has been widely regarded as the primary driver of socio-economic progress and prosperity. Consequently, it has become the focus of numerous studies across various social research disciplines, giving rise to a substantial research industry (Godin & Vinck, 2017) dominated by economic sciences and business management with specific emphasis on economic growth (Godin, 2020; Lundvall, 2013, p. 35). While innovation is widely acknowledged, even within business and economic literature, as a primarily social phenomenon and a process shaped by social forces (Mytelka & Smith, 2002; OECD, 1992; Pansera & Fressoli, 2021; Pratt, 2017; Ruzzeddu & Ferone, 2018) it has not come withing the focus of sociological research. Instead, it has been considered as a part of other research subfields such as various technology assessment studies (TA) focused on critical approach to new technologies; social studies of science (SSS) mainly in the context of actor-network theory (ANT) (Latour & Woolgar, 1979) and the theory of social construction of technology (SCOT) (Pinch & Bijker, 1984);

innovation studies (IS) in the early stage of its development (Fagerberg & Verspagen, 2009; Martin, 2016, p. 439) and finally of science and technology studies (STS) (Law, 2008, p. 634; Martin, 2012).

Attempts to establish the sociology of innovation as a distinct research subdiscipline did not fall on fertile ground (Brandão & Bagattolli, 2023; Esquinas, 2021; Knoblauch, 2018; Martin, 2012; Martin et al., 2012). Knoblauch, for example, conclude that '(D)espite its relevance to the economy and to knowledge and technology management, innovation has not become a central category of sociology or other basic research' (Knoblauch, 2018, p. 111). Esquinas (2021) acknowledged the importance of sociologists like Ogburn (1992), Merton (1968), Mulkey (1979/2015) or Collins (1983) in understanding innovation triggered by science and technology but stated that in recent years, the contribution of sociology has been sporadic and unsystematic. Moreover, he holds that the sociology of innovation as a distinctive specialty is inconspicuous in current scientific production and academic programmes. Brandão and Bagattolli (2023, p. 61) observed that critical sociological studies of innovation as a systemic phenomenon in our societies, encompassing its unintended and adverse outcomes, have emerged relatively recently, around 2010. These studies predominantly draw upon the research of B. Godin (2015, 2020), a prominent figure in investigating the history of innovations who is recognized as one of the pioneers in promoting a novel agenda: critical studies of innovation.

Hence, it can be stated that, apart from the foundational concepts within the sociology of innovation, such as G. Tarde's imitation of innovation (Tarde, 1903) and E. M. Rogers's (1962) diffusion of innovation, sociology's contribution to the study of innovation has remained sporadic and lacking in systematic development. Social sciences, particularly sociology, have shown a greater inclination towards examining various risks associated with modern societies (Beck, 1992) such as globalization (Giddens, 2000), platformization (Srniczek, 2016; van Dijck et al., 2018), scientification (Bell, 1973; Böhme & Stehr, 1986), digitalization (Betancourt, 2015) and the concept of liquid modernity (Bauman, 2000) among others.

STS is widely acknowledged as the 'true home' for the sociology of innovation since both are focused on interaction between science, technology and society and share common origins in the sociology of science and scientific knowledge. Scholars (Esquinas, 2021; Law, 2008; Martin, 2012) clearly categorized sociologists of innovation as integral participants in STS studies. Sociologists, alongside historians, philosophers and anthropologists, share the same principles concerning the role of knowledge in society and exhibit corresponding cognitive affinities. Unlike STS, which became professionalized in the 1970s (Martin et al., 2012, p. 1192) sociology of innovation did not define its central theoretical and methodological concepts, neither did it organise educational programmes, career paths for scholars or forums for sharing research which are all important for institutionalisation of any research discipline.

This research endeavours to explore whether it is possible today, in the present era marked by global societal challenges encapsulated in the United Nations' Sustainable Development Goals (UN, 2015), and by socially disruptive technologies (Hopster, 2021), to rethink the sociology of innovation as an independent research subdiscipline. The research is based on the strong belief that the sociology of innovation represents an evolving and dynamic scientific field whose expansion and autonomy is supported by a recent paradigmatic shift in our understanding of innovation. This shift advocates for the separation of innovation from its previous sole focus on economic growth and business competitiveness (Brandão & Bagattolli, 2023; Pansera & Fressoli, 2021) and

instead advocates for a new innovation paradigm where innovation serves the purpose of socio-technical transitioning to a more sustainable and humane society (Geels, 2020).

Consequently, the main motive of this research is to encourage discourse surrounding the sociology of innovation as a significant sub-discipline within sociology. As a result, it should aim to develop analytical and theoretical frameworks, educational programmes, and other discipline-specific initiatives aimed at catalysing transformative social change through innovation. The ultimate goal is to establish the sociology of innovation as an influential actor in guiding innovation development for the transformation of socio-economic systems, aligned with sustainable and inclusive growth and well-being.

Following this motive, the objective of this research is to identify the theoretical foundations and intellectual underpinnings of the sociology of innovation, ultimately drawing conclusions about its future research prospects and directions.

The main research questions are:

- How has innovation evolved to one of the main determinants of society and therefore of interest for sociology?
- What are the intellectual basis and theoretical and conceptual roots of sociology of innovation?
- What are the research subject and tasks of sociology of innovation within socio-technical transition?

The evolution of the concept and paradigm of innovation pertaining to the first research question and outlined in the Section 3, significantly determines the discourse, subject and goals of the social studies of innovation, including sociological inquiries. The historical trajectory shows how innovation has come a long way from socially deviant behaviour (Merton, 1968), through being the main driver of the knowledge society (Bell, 1973; Böhme & Stehr, 1986; Fagerberg, Landström, et al., 2012; Godin & Vinck, 2017) towards the current scepticism about innovation as a panacea for solving socio-economic problems (Godin, 2015, 2020; Godin & Vinck, 2017). Presently, innovation is expected to actively engage in socio-technological transitions (Geels, 2002, 2020) and to contribute to transformative innovation policies aimed at achieving desired societal objectives (Mazzucato, 2021; Schot & Steinmueller, 2018; Weber & Rohracher, 2012).

This evolutionary path towards human-centered innovation is supported by the analysis of the theoretical and conceptual roots of sociology of innovation, specifically addressing the second research question and further elaborated upon in Section 4. It describes the intellectual base upon which the sociology of innovation is built, illuminating how scholars formulate their theories and discourses over time regarding the nature of innovation, their roles, and the resulting social consequences.

The motivation for conducting this analysis stems from the observation that while the intellectual foundations of neighbouring disciplines, particularly innovation studies (IS) (Fagerberg, Fosaas, et al., 2012; Fagerberg & Verspagen, 2009), science and technology studies (STS) (Martin, 2012; Martin et al., 2012), entrepreneurship studies (Landström et al., 2012), and knowledge society studies (Fagerberg, Landström, et al., 2012), have been extensively examined in the referenced works, the overview of the research disciplines comprising the knowledge base of the sociology of innovation remains insufficiently explored and lacking systematic analysis. Therefore, an original contribution of

this study is an integrative literature review of the evolution of theories and concepts developed within relevant research fields and neighbouring disciplines that shape the current sociological discourse on innovation.

The roots of the sociology of innovation are traced, as outlined in Table 1, across four frameworks that initially emerged from classical sociology of knowledge, subsequently

Table 1. Knowledge foundations of the sociology of innovation

FUNDAMENTAL THEORETICAL FRAMEWORKS		MAIN MESSAGE
1. CLASSICAL SOCIOLOGY OF KNOWLEDGE		
Marx, K. (1867)	<i>Das Kapital</i>	Human ideation (knowledge, ideas, ideologies) develops within the social context of a society's groups and institutions. Scientific truths are exempted.
Weber, M. (1904-1905/2001)	<i>The Protestant Ethic and the Spirit of Capitalism</i>	
Durkheim, E. (1912/1995)	<i>The Elementary Forms of the Religious Life</i>	
Mannheim, Karl (1928/1960)	<i>Ideology and Utopia: An Introduction to the Sociology of Knowledge</i>	
2.a. SOCIOLOGY OF SCIENCE		
Merton, R. K. (1968)	<i>Social Theory and Social Structure</i>	Science institutions and orgaiaations are socially conditioned. Science does not evolve gradually but through paradigm shifts (revolutions)
Kuhn, T.S. (1962)	<i>The Structure of Scientific Revolutions</i>	
2.b. SOCIOLOGY OF SCIENTIFIC KNOWLEDGE (SSK)		
Bloor, D. (1991)	<i>Knowledge and social imagery</i>	Not only science organization but also the content of scientific knowledge (including 'hard' sciences) are socially influenced; the cognitive is the social and vice versa
Collins, H.M. (1983)	<i>The sociology of scientific knowledge: Studies of contemporary science (article)</i>	
Mulkay, M. (1979/2015)	<i>Science and the sociology of knowledge</i>	
3. SCIENCE AND TECHNOLOGY STUDIES (STS)		
Pinch, T. J., Bijker, W. E. (1984).	<i>The Social Construction of Facts and Artefacts: or How the Sociology of Science and the Sociology of Technology might Benefit Each Other</i>	Technology should be separated from science; not only science, but technology is socially constructed; Sociology is just one ingredient in formation of STS
Latour and Woolgar (1979)	<i>Laboratory Life: The Construction of Scientific Facts (ANT)</i>	
4. INNOVATION STUDIES (IS)		
Schumpeter, J. A. (1943)	<i>Capitalism, Socialism, and Democracy</i>	Innovation is a driver of capitalism; innovation and socio-economic progress are synonymous; innovation is an interactive (not linear) and evolutionary phenomenon; creation of innovation requires an appropriate institutional support (national innovation system) and deliberate policy actions; recent societal global challenges require transformative change in understanding innovation and innovation policy for the betterment of society
Nelson and Winter (1982)	<i>An Evolutionary Theory of Economic Change</i>	
Lundvall, B.A. (1992)	<i>National system of innovations, towards theory of innovation and interactive learning</i>	
Freeman, C. (1987)	<i>Technology, policy and economic performance: Lessons from Japan.</i>	
Weber and Rohrer (2012)	<i>Legitimizing research, technology and innovation policies for transformative change</i>	
Schot and Steinmueller (2018).	<i>Three frames for innovation policy: R&D, systems of innovation and transformative change</i>	
5. SOCIOLOGY OF INNOVATION		
5.1. Sociology of diffusion		
Tarde, G. (1903)	<i>The Laws of Imitation</i>	Innovations are spread by imitation, social interaction, and communication
Rogers, E. M. (1962)	<i>The Diffusion of Innovations</i>	
5.2. Triple helix model (THM)		
Leydesdorff and Etzkowitz (1998)	<i>The Triple Helix as a model for innovation studies</i>	Science-industry cooperation is a driver of innovation and economic growth; entrepreneurial universities challenge traditional universities
Etzkowitz (2008)	<i>The Triple Helix: University-Industry-Government Innovation in Action</i>	
5.3. Multilevel perspective (MLP)		
Geels, F. (2002, 2020)	<i>Technological Transitions as Evolutionary Reconfiguration Processes: A Multi-Level Perspective and a Case-Study</i>	Innovation can be understood through multilevel perspective (landscapes, regimes and niches); There is a need for socio-technical transitions towards sustainability

extending into the domains of the sociology of science and the sociology of scientific knowledge. In the beginning of the twentieth century, the concept of innovation was detached from science as its intellectual base, becoming a productive technological force. This transition led to the emergence of the third theoretical and analytical framework within innovation research, notably in the field of science and technology studies (STS), which drew inspiration from Kuhn's work (1962) and social constructivism (Latour & Woolgar, 1979; Pinch & Bijker, 1984). The fourth framework refers to the innovation studies (IS) rooted in the ideas of Schumpeter (1943) and dominated by evolutionary and institutional economics (Edquist, 1997; Freeman, 1987; Nelson & Winter, 1982) focused on institutional setups for fostering innovation as driver of business competitiveness.

The development of the concept of innovation in a narrower sociological sense began with seminal ideas put forth by authors such as Tarde (1903), Rogers (1962), Ogburn (1922), and Gilfillan (1935a, 1935b). However, these concepts were not widely popularised within sociological research. Current sociological inquiries into innovation primarily revolve around the Triple Helix Model (Leydesdorff & Etzkowitz, 1998), which explains how academia, industry, and government interact to promote technological innovation for economic growth. However, this research, premised on the notion that the sociology of innovation should catalyse social changes through innovations that address both economic growth and persistent societal challenges in modern societies, draws inspiration from the perspective of socio-technical transitions as introduced by F. Geels (2002, 2004). This perspective is fundamentally rooted in sociological discourse, highlighting the complex interplay among technological innovations, social structures, and institutional frameworks in propelling radical transformations towards sustainable and more humane societies.

Such a long-term strategic processes oriented to system transformation lie at the heart of transformative innovation policy (Fagerberg, 2018; Schot & Steinmueller, 2018; Weber & Rohrer, 2012). This policy emerges in response to the growing recognition that IS and STS likely face limitations in effectively addressing urgent societal issues such as environmental degradation, climate change, and socio-economic disparities (Soete, 2019; Steinmueller, 2013). These challenges have prompted numerous scholars to recognize the adverse socio-economic impacts of innovation (its 'dark side') and advocate for a 'normative turn' in understanding innovation (Biggi & Giuliani, 2020; Coad et al., 2021; Diercks et al., 2019; Soete, 2019). Innovation scholars are increasingly giving prominence to the rise of transformative innovation policies as a crucial facilitator of the 'social turn' necessary for the transition towards a more sustainable and humane society (Geels, 2020; Geels, 2020; Haddad et al., 2022; Hekkert et al., 2020; Lundvall, 2024; Martin, 2016; Mazzucato, 2021; Schot & Steinmueller, 2018; Soete, 2019; Walsh et al., 2020; Weber & Rohrer, 2012).

Following this path, the legitimate question in the current changing socio-economic context (the third research question of this research) is whether sociology can offer proper concepts, theories and normative orientation to innovation which is expected to enable both wealth, and prosperity on the one side and sustainability and humanity in the other side. Can it help concrete actions like European climate neutrality by 2050 (EC, 2019), transition to electrical vehicles, or even facilitate deeper socio-economic transition towards a fairer and more equal society to remedy the retreat of liberal

democracy (Wolf, 2023, p. 63)? Is it possible to encourage formation of the sociology of innovation beyond the bounds of STS, as a sub-discipline of sociology with its specific subject of research, theory and methodology and disciplinary determinants like professional associations, curricula, journals, or conferences?

This research, in addressing the aforementioned questions, relies on the ongoing debate about sociology's role in society, initiated by Burawoy's (2005) call for a more proactive, public-oriented sociology to regain its diminished social relevance and visibility. Aligning with Burawoy's concepts of organic public sociology, which prioritize the interests of civil society, this research proposes that the sociology of innovation should produce scientific knowledge to foster human-centred innovation capable of catalyzing social change in alignment with civil society's interests and the advancement of social welfare.

This research makes a unique contribution in several ways. Firstly, it offers a comprehensive overview of the historical development of ideas and concepts that form the intellectual foundation for the sociology of innovation, filling a gap in the existing literature. Secondly, it illustrates how the evolution of these concepts, particularly within the domains of IS, STS, innovation policies, and transition studies, leads to a shared understanding of the necessity to reconceptualize innovation as a catalyst for socio-technical transition toward a humane and sustainable society. Thirdly, the research suggests that this perceived necessity for socio-economic transformation creates opportunities for the development of sociology of innovation distinct from IS and STS. Finally, the research proposes that the objectives of the sociology of innovation should adhere to the original sociological discourse, exemplified by Burawoy's concept of organic public sociology. Such a sociological approach involves theoretical, analytical and normative exploration of innovation to generate social change aligned with the welfare of society, reflecting authentic sociological concerns.

The methodology employed in this research is explained in the following section. The third section offers a brief description of the evolution of the concept of innovation, emphasizing its role as a key determinant of socio-economic progress. In the fourth section, the focus shifts to identifying the knowledge base of the sociology of innovation and its historical embeddedness in neighboring disciplines. The implications of this research are discussed in the fifth section, paving the way for a subsequent discussion on the potential future development of the sociology of innovation in the sixth section. Finally, the seventh section summarizes the main conclusions derived from the study.

2. Methodology

This research examines the feasibility of advancing the sociological discourse of innovation as an independent sub-discipline within sociology, distinct from IS and STS, considered as its primary domains. Starting with the premise that the evolution of theories and concepts across relevant research fields and neighboring disciplines shapes the sociological discourse on innovation, this study aims to trace the historical evolution of the intellectual base of sociology of innovation.

Since the contemporary understanding of innovation as a productive force emerged only in the twentieth century, the review begins by examining the classical sociology of knowledge, which lays the initial intellectual basis for comprehending innovation. It

culminates with the concept of the sociology of innovation as a potent agent of societal transformation. This discourse draws upon Burawoy's concept of organic public sociology, integrating it with the recent conceptions of sociotechnical transition and transformative innovation policy.

The process of identifying key concepts from various research fields to elucidate their impact on sociological research of innovation relies on a critical qualitative analysis of literature sources. Given the absence of established theoretical and analytical grounds for the sociology of innovation, this research draws on existing literature within four key theoretical frameworks: classical sociology of knowledge, sociology of science and sociology of scientific knowledge, innovation studies (IS), and science and technology studies (STS). In addition, the evolution of the innovation concept within a more specific sociological context encompasses the sociology of diffusion, the Triple helix model, and the perspective of socio-technical transition.

The literature across these frameworks is extensive, dispersed, and fragmented, rendering a literature review a demanding task. This condition makes the full systematic review process difficult, potentially resulting in an inability to provide a coherent and comprehensible 'state of the field' (Snyder, 2019). To avoid these pitfalls, the literature review is based on the integrative literature review approach, which differs in scope and purpose compared to systemic and narrative literature reviews, which are the most basic types of literature reviews (Toronto, 2020).

The goal of an integrative review is to look more broadly at the topic of interest than a systemic review, aiming to develop a holistic understanding of the phenomena. Integrative literature reviews typically focus on two types of topics – mature topics and new, emerging topics (Snyder, 2019; Torraco, 2005, 2016). With mature topics, characterized by a broad and diverse knowledge base, the integrative review seeks to comprehend the subject by synthesizing diverse sources to produce a new model, conceptual framework, or other unique conception informed by the author's expertise on the subject (Torraco, 2005, 2016). On the other hand, for new topics lacking theoretical and conceptual foundations in broad literature, an integrative review is more likely to result in an initial or preliminary conceptualization of the topic, such as a new model or framework (Torraco, 2005, 2016).

An integrative review typically aims to evaluate, critique, and synthesize literature on a research topic in a way that enables the emergence of new theoretical frameworks and perspectives (Snyder, 2019; Torraco, 2005, 2016). Therefore, the integrative review stands out as the most effective method for critically reviewing literature in this research, leading to a preliminary conceptualization of the sociology of innovation as a new topic. This methodological approach enables the analysis of diverse strands of thought, narratives, and theoretical approaches regarding the evolution of the concept of innovation, thus laying the intellectual foundation for the sociology of innovation as an emerging concept within this research.

3. Evolution of the concept of innovation

The concept of innovation has evolved significantly over time, reflecting changes in historical and sociocultural contexts which shape our understanding of how and why innovation takes place. Despite a vast body of literature, it is very difficult to provide a

comprehensive definition of the term since innovation is today a multidimensional ‘umbrella’ concept that includes various meanings and definitions from the perspective of different disciplines (Godin, 2015). As such innovation can be entrepreneurial (Autio et al., 2014), scientific (Nelson, 2004), open (Chesbrough & Bogers, 2014), collaborative (Blomqvist & Levy, 2006), disruptive (Christensen, 1997), green (Taklo et al., 2021), dark (Coad et al., 2021), sustainable (Afeltra et al., 2023), or transformative (Weber & Rohra-cher, 2012), reflecting diverse disciplinary approaches to evolution of the concept. The rise of digital innovation, which is increasingly ethereal, disembodied, and intangible (Švarc, 2022) challenged the conventional concept of technological innovation which advances industry and has led some scholars to conclude that innovation today means anything, everything, and nothing (Godin, 2015, p. 8; Taatila et al., 2006).

However, innovation has come a long way to become a factor of progress and growth. The history of the development of innovation has shown that innovation had nothing in common with creativity and technology until the twentieth century (Godin, 2015; Godin, 2020; Godin & Vinck, 2017). It was mainly a politically derogative label for opponents and enemies and a term that was usually contested and pejorative since it brought about changes to the established social order and therefore was dangerous and subversive.

For instance, R. Merton (1968, p. 230) understood innovation as socially deviant behavior (along with ritualism, retreatism and rebellion) aimed at achieving culturally recognized goals such as ‘American Dream’. However, it can be tolerated to a certain degree since innovation may result in the formation of new institutionalized patterns of behavior which are more productive for the realization of desired goals.

Apart from some earlier contributions by Joseph Schumpeter (1911/1934), Gabriel Tarde (1903), and Everett M. Rogers (1962), there was no explicit use of the concept of innovation before the Second World War. During this period, military needs drove many technological innovations that found applications in various science-based industries, including chemical, nuclear, computer, aviation, pharmaceuticals, and more. This marked a reversal in the perception of innovation, turning it into a useful and creative activity that became a progressive economic force (Fagerberg, 2016; Godin, 2015).

Scholarly interest in studying innovation exploded in 1960s with the rise of innovation studies (IS) (Fagerberg & Verspagen, 2009; Fagerberg et al., 2012; Martin, 2012, 2016), when it became clear that the position of any country (mostly America at that time) rested on technological supremacy based on R&D embedded in innovation (Fagerberg & Verspagen, 2009; Fagerberg, Fosaas, et al., 2012; Freeman, 1987; Godin, 2015; Mytelka & Smith, 2002). In industrial societies innovation is usually understood as the development of new ideas into marketable products and processes or the first application of science and technology in a new way, with commercial successes (OECD, 1971, p.11).

This view culminated in the 1990s with the rise of the ‘knowledge society’ (Bell, 1973; Böhme & Stehr, 1986; Fagerberg, Landström, et al., 2012) when scientific and technological advances transformed into commercially viable innovations were seen as drivers of long-term economic growth, employment, and social welfare (Godin & Vinck, 2017; OECD, 1971). Empirically, economic statistics have evidenced that investment in knowledge has grown more rapidly than investment in machinery and equipment since the mid-1990s in most OECD countries (OECD, 2005a, 2005b). This promoted innovation into an uncontested value (Godin & Vinck, 2017, p. 5), a basic concept of development

policy, which makes a shift of science policy to become innovation policy (Fagerberg, 2016).

Innovation policy spread around the globe and has become a central element in the attempt of the European Union (EU) to strengthen its economy since the mid-1990s when it faced slow economic growth and lower competitiveness in the world market (Fagerberg & Verspagen, 2009). European innovation policy has gone through several phases in understanding of the role of innovation and policy models for its support. Firstly, innovation was understood as a solution to the European Research Paradox (European Commission, 1995), then as a tool for transformation of the European Union into a highly competitive and knowledge-based economy by 2010 within the Lisbon agenda (European Commission, 2000), and most recently, innovation has become the basic concept of the smart specialization strategy (S3) which is currently widely exercised in the EU. In this phase, support for innovation moved away from R&D towards the entrepreneurial discovery process and the needs of regional development (Foray, 2014).

In parallel with policy evolution, there is also a reconceptualization of the original concept of innovation, traditionally measured through R&D inputs as defined in the Frascati manual (OECD, 2002). The new concept of innovation has significantly broadened, extending beyond technological advancements to include non-technological aspects such as service, organizational, and marketing innovations. This broader perspective was first outlined in the 2005 *Oslo Manual* (OECD, 2005a, 2005b, p. 46) and further modernized in the 2018 edition (OECD/EUROSTAT, 2018, p. 4), to better capture the role of digital transformation and service economies, as well as social and business model innovations. This change is driven by the increasing trend of deindustrialization and the emergence of the creative class (Florida, 2002) mainly consisting of ‘knowledge workers’ which shift away from traditional industry-based economies. Innovation has evolved into a soft concept that covers almost every novelty in the domain of organizational, marketing and business innovation which come up with market success. Despite this significant shift in understanding innovation, it has not incorporated a ‘normative turn’ addressing the broader ethical, societal, and environmental implications of innovation.

Today, the concept of innovation is once again facing great challenges which require a paradigm shift and a ‘normative turn’ of our understanding of innovation, (Biggi & Giuliani, 2020; Diercks et al., 2019; Soete, 2019; Uyarra et al., 2019). This turn is crucial for overcoming the potential negative consequences of innovation, often referred to as the dark side of innovation (Coad et al., 2021), which can be harmful to both society and nature. Innovation should be decoupled from maximization of economic growth (Pansera & Fressoli, 2021), while the role of innovation should extend beyond merely enhancing economic growth; it should guide processes of transformative change towards desired societal objectives.

4. Knowledge base for the sociology of innovation

This section aims to systemize and summarize the main directions in social sciences which contribute to the sociology of innovation. The purpose is to identify main concepts from different research fields to understand how they made influence on sociological research of innovation, rather than provide a systemic literature review or details of influence. Since any history is an artful construction (Law, 2008, p. 624), this systematization

is inevitably arbitrary but offers a concise insight into the history and antecedents of the sociology of innovation.

Considering that the modern concept of innovation, as we understand it today in terms of technological and marketable new ideas, did not exist before the twentieth century (Godin, 2015), the sociology of innovation initially traces its origins to classical sociology of knowledge and subsequently extends into the realms of the sociology of science, scientific knowledge, and technology. Studying innovation ‘through’ science is rather natural since there is the difficulty to separate innovation from its intellectual base which relies heavily on the creation of basic knowledge through both education and science. Therefore, Pratt (2017, p. 231) stated that our common understanding of innovation comes from science. Scientific discovery is practically identified with innovation, especially during the era of scientific and technological revolution, in the late nineteenth and early twentieth centuries, when the process of the conversion of science into productive force started.

Tracing the historical development of the intellectual foundations of the concept of innovation, from its origins in scientific knowledge to its contemporary interpretation, four key theoretical frameworks emerge as significant contributors to the sociology of innovation. These frameworks represent the stages in understanding of knowledge over time, as both a subject of study and its role within the social context, often shaped by critiques of earlier theories and discourses.

- Classical Sociology of Knowledge;
- Sociology of Science and Sociology of Scientific Knowledge (SSK);
- Science and Technology Studies (STS);
- Innovation Studies (IS).

Despite drawing on these broader theories, the sociology of innovation has followed a distinct developmental path since the early twentieth century. Within this specialized context, three specific subfields have emerged:

- Diffusion theory;
- The Triple Helix Model;
- Socio-Technical Transformations.

The next section will explore these theoretical frameworks to grasp how scholars from diverse backgrounds contribute to shaping the evolution of the sociology of innovation and offer fresh perspectives.

4.1. Classical sociology of knowledge

The central theme of classical sociology of knowledge was the exploration of the relationship between knowledge, broadly understood as human ideation, and the larger society with the aim to clarify the interplay between ‘existence and thought’ (Eisenstadt, 1987, p. 77). It flourished in the 1920s and 1930s, after the First World War, at a time when economic and political conflicts begin to be reflected in the sphere of thought. Conflicting worldviews and interpretations of world raised interest of scholars for ‘social origins’, or

‘realities’ of ideas and opinions. Research was framed by the works of Karl Marx, Max Weber and Emile Durkheim and above all by Karl Mannheim, a German sociologist and philosopher (Eisenstadt, 1987; Hamilton, 1974; Stehr & Meja, 1982). Although Marx, Weber and Durkheim did not explicitly discuss science and innovation as a topic for substantive sociological study, there are elements in their theories which influence conceptualization of innovation to today.

Marx’s ‘great contribution was that he brought out for the first time the basic social character of science and the corresponding necessity of science to society’ (Bernal, 1952, p. 5). Starting from the Hegelian dialectic and idealistic concept of history which takes place in the speculative world of ideas, Marx developed historical materialism as a general theory and a method of social development (Marx, 1867). Human history no longer takes place through the development of ideas in the abstract theoretical world, but in a real world determined by economic activity and labor captured in production forces and production relationships. In simpler terms: man’s consciousness is determined by his social being.

In contrast to Marx, Max Weber argued that development of capitalism is not fueled only by material conditions but also requires ideological and spiritual formulations (Hamilton, 1974) that he found in the protestant ethic as a particular form of rationalism (Weber, 1904-1905/2001). Religion provides rational justification for individuals and social groups to be economically active and for capitalistic regime to develop. Socio-historical change, to Weber, is moving towards a greater rationalization of human activity, or greater technical-purposive control (instrumental action ideal type) over nature, society, and culture (Hamilton, 1974). This theory is also relevant today for understanding the unceasing progress of knowledge and exploitation of innovation.

While Marx’s sociology of knowledge is framed by materialism and productive forces, and Weber’s by individualism and rationalism, Durkheim’s substantial theory is concerned with collectivism and relativism. Durkheim (1912/1995) suggests that social life is possible only through the collective consensus around the basic ideas and concepts that arise from the collective mind which emanates a collective representation. People’s beliefs, norms and values make up a collective consciousness (representations), or a shared way of understanding and behaving which creates social integration. Truth, in this context, is a human creation, a vision filtered through human interests and specific social arrangements (Gieryn, 1982) while logic is no less a product of society than, for example, ethics. In this vein, Durkheim’s substantive theory points in the direction of a kind of ‘cognitive relativism’ (Gieryn, str. 108) since the validity of truth is not universal but relative to designated historical contexts, social structures and processes and grounded in collective mind. However, when he considers modern science, Durkheim deviates from his general idea of relativism of human cognition and holds that intellectual activities are increasingly liberated from social constraints through the evolution from traditional to modern society (Collins & Restivo, 1983). More complex societies, in contrast to small-scale societies permeated by organic solidarity, become less restricting allowing scientific concepts to be accepted because they are true and objective, not because they stem from collective representants (Gieryn, 1982).

Although contributions of Marx, Weber and Durkheim were considerable, scholars mainly agree that K. Mannheim established sociology of knowledge (Eisenstadt, 1987; Hamilton, 1974; Matthew, 2005; Mulkay, 1979/2015; Stehr & Meja, 1982) in his book

‘Ideology and Utopia’ (Mannheim, 1928/1960), by his theory that all human knowledge is related to existence (existentially determined) and depending on the social position and perspective of a particular society or group. In contrast with Marx, who perceives ideology as a distorted consciousness of certain social group, Mannheim holds that all human knowledge and truth, including sociology of knowledge, is a certain type of ideology which reflects its social context. Exceptions are natural sciences and mathematics, since relationships in the natural world are timeless and universal and independent of culture and social position. He coined the term ‘relationism’ (in contradistinction to ‘relativism’) to denote the epistemological perspective of his sociology of knowledge (Hamilton, 1974, p. 127/128; Mulkay, 1979/2015, pp. 12–14).

The classical sociology of knowledge has enhanced our understanding of how social structures, institutions, and relationships influence the creation and dissemination of human cognition. These perspectives have played a foundational role in shaping contemporary sociological approaches to the study of both knowledge and innovation, as well as their interconnection with society.

4.2. Sociology of science and sociology of scientific knowledge

While classical sociology of knowledge portrayed science as an esoteric activity having its own internal logic which is only of marginal interest for sociology (Mulkay, 1980), the definite separation of cognitive and social processes in science was made by R. Merton (1968), the American sociologist who established the sociology of science as Mannheim established the sociology of knowledge. Merton and other functionalist sociologists (e.g. Barber, 1952) were establishing their intellectual dominance in American academia during the 1940s and 1950s (Collins, 1983; Eisenstadt, 1987) and highlighted that science serves as a social function of providing certified knowledge needed for scientific progress. As Merton’s science is not exclusively the result of intrinsic scientific logic but the result of professional scientific communities, his research was focused on social conditions, institutional structures, and processes which enable scientific communities to generate and evaluate scientific claims (Ben Ben-David & Sullivan, 1975; Martin et al., 2012). That requires the norms of universalism, disinterestedness, communalism and organized skepticism which structure the social interactions of the scientific community and enable scientific activities (Martin et al., 2012; Matthew, 2005).

According to Collins (1983), the take-off of the sociology of science into a flourishing research area occurred in the early 1960s, with works by Derek de Solla Price (1963) and Thomas Kuhn (1962), as well as important studies by Joseph Ben-David (1960) and Warren Hagstrom (1965). The greatest influence was made by Kuhn (1962) whose theory departed from the Mertonian tradition. He established the idea of the scientific paradigm as an alternative epistemological assumption (Law, 2008) that interconnects knowledge, methods and social aspects of science and establishes paradigm shifts in science development instead of its gradual evolution. In contrast to traditional sociology, which often displays ignorance regarding actual scientific practice, Kuhn’s theory of alternating between periods of normal and revolutionary science aligns much more closely with real scientific practices. It is widely recognized and appreciated by many researchers, leading to its global popularity (Law, 2008; Matthew, 2005).

Although the sociology of science treated science as a social activity, the attribute ‘social’ was used in a very limited sense, only to describe circumstances that are external to the intellectual activity of science itself (structural-organizational components of science, norms and values, etc.) while scientific knowledge – its structure and content – remained outside the scope of sociology and beyond social forces and social criticism (Ziman, 1989). The interest of sociologists in the cognitive side of science – the content of its claims and theories – emerged in the early 1970s within the sociology of scientific knowledge (SSK) and successfully replaced the Mertonian limited focus on scientific institutions (Matthew, 2005). In contrast, SSK scholars hold that the cognitive is the social and vice versa, and that everything can be analyzed in terms of discourse (Leydesdorff, 1992). SSK is usually identified with the Strong Programme (strong in the sense of strong epistemological relativism) spearheaded by David Bloor (1991) at the University of Edinburgh during the 1970s and taken up by others like H. Collins (1983) (‘Bath school’) and Mulkay (1979/2015) at the university of York (Collins, 1983; Matthew, 2005; Shapin, 1995; Ziman, 1989).

The Strong Programme appeared as reaction to the Mertonian ‘sociology of error’ (Collins, 1983; Haddock, 2004) or ‘weak programme’ in sociology of science by which only scientific irrationalities or errors should be sociologically explained while true science is sufficiently explained by the presence of the real world (Haddock, 2004; Kemp, 2005). Bloor holds that knowledge consists of collectively endorsed ‘beliefs which people confidently hold to and live by’ (Bloor, 1991, p. 51). He states that sociologists should follow the principles of symmetry and investigate the origins of all kinds of knowledge including ‘hard sciences’ (Pinch & Bijker, 1984, p. 401; Sismondo, 2010). Bloor’s approach made a turn in the sociology of knowledge, stating that not only science organization but also the content of scientific knowledge is socially influenced, making the Strong Programme the first and most prominent social constructionist approach to scientific knowledge (Haddock, 2004; Kemp, 2005).

The Sociology of Science and the Sociology of Scientific Knowledge (SSK) contribute to the sociology of innovation by explaining that innovation is not merely the outcome of objective, neutral, and merit-based processes. Instead, it emerges from the accumulation of knowledge shaped by specific scientific or technological paradigms and is significantly influenced by cultural and institutional norms, ethics, social acceptance, and political discourse.

4.3. Science and technology studies (STS)

Science and technology studies (STS) emerged in the 1970s as a distinct interdisciplinary field – focused not only on social aspects of science but also on the social and political implications of technology as a distinctive branch of knowledge. The dramatic growth of science as described by Derek de Solla Price (1963), which made science a major factor of industrial production and economy after the Second World War, blurred the boundaries between science and technology, making technological practices themselves become an interesting research area (Sismondo, 2010). Scholars became interested in the relationship between scientific knowledge, technological systems, and society delving into questions about the impact of technology and technology policy on social relations, public goals, humanity, freedom and other related aspects (Jasanoff, 2011; Sismondo, 2010).

STS also emerged as a response to theories of technological determinism advocated by certain authors, such as W. Ogburn (Godin, 2010). These deterministic theories posit that technology functions as an external, autonomous force steering the development of social structures, limiting the possibilities for democratic engagement with technology.

The foundational roots of STS can be traced back to Kuhn's original perspective on scientific revolutions and the constructivist approach of the Strong Programme (Jasanoff, 2011; Law, 2008; Martin et al., 2012; Sismondo, 2010). However, the beginning of STS is marked by the extension of SSK idea of social construction (a term borrowed from Berger & Luckmann, 1966) of scientific truths to study of technological artefacts (Law, 2008; Williams & Edge, 1996). Basically, two theories have made a 'technological turn' towards sociology of technology, both based on the idea that technology, similarly to science, is socially constructed (Martin et al., 2012). The one theory is the 'Social construction of technology' (SCOT) by Pinch and Bijker (Pinch & Bijker, 1984; Bijker et al., 1987), and other is the Latour and Woolgar's (1979) Actor-network theory (ANT).

Both theories claim that social and the technological phenomena are not distinct spheres, but one and the same. Pinch and Bijker argued that the success or failure of certain technologies (e.g. the bicycle) is socially constructed through the interpretation of relevant social groups, influencing the alternation and variation of technology, making it a 'multi-directional' rather than a linear phenomenon (Pinch & Bijker, 1984, p. 411). However, they faced criticism for overemphasizing social choice over technological considerations and were accused of social determinism.

Latour and Woolgar (1979) and Callon (1990) went a step further in overcoming both technological and social determinism by presenting science-technology knowledge (techno-science) as the creation of relation within networks of human and non-human actors, called 'actants' that include everything which might have impact on scientific practice (microbes, institutions, organizations, machines). This suggests that science and technology are products of interaction between a large number of diverse actors with no distinction between the social and technical elements of science. Both SCOT and ANT were mainly interested in micro-level research of how science become technological artefacts (Jasanoff, 2011) and were criticized for lack of consideration of wider social structures (macro level) and for political and moral indifferences. A lack of the economic, political and ideological influences on establishing networks and choice of technologies as well as the sociological strict distinction between 'the natural' and 'the social' distanced sociology from ANT and further structuralist approach. Nevertheless, contemporary social research continues to draw inspiration from ANT theory, giving rise to a 'second generation' of ANT scholars who are working to extend ANT principles to the proliferation of modern technologies (Blok et al., 2020).

From the 1960s onwards, the STS community grew in size and geographical coverage, and from the 1970s STS was becoming professionalized with the formation of bodies such as the Society for Social Studies of Science (founded in 1975) and the European Association for Studies of Science and Technology – EASST (founded in 1981), as well as the creation of specialist STS journals, in particular Social Studies of Science (SSS, established in 1971), and Science, Technology and Human Values (set up in 1976) (Martin et al., 2012).

STS, as an interdisciplinary field, will remain crucial in shaping our future due to its strong emphasis on the responsible and balanced development of science and

technology. It will continue to serve as a fertile ground for theoretical and empirical exploration of the socio-cultural, political, and ethical dimensions of science and technology, especially in the face of today's numerous global challenges (Konrad et al., 2016).

On the other hand, STS has developed since its inception in 1970s, in myriad topics, and its variety has been proliferating to such an extent that there is hardly any place where STS cannot find themes for study (Matsumoto, 2010). STS received many criticisms for its fragmentation and lack of scientific rigor, and has become a rather divided community (Martin et al., 2012, p. 1194). This paves the way for the sociology of innovation to detach itself from being one of the numerous and fragmented components of STS, and to define its specific body of knowledge and understanding what socio-economic goals this knowledge should serve. This naturally begs the question of what is the social, analytical, and normative focus of sociology of innovation?

4.4. Innovation studies

Innovation studies (IS) is a relatively young field of research which did not, with the exception of Schumpeter (1911/1934/1911) and others, exist before the late 1950s (Fagerberg & Verspagen, 2009; Fagerberg, Fosaas, et al., 2012; Fagerberg, Landström, et al., 2012; Martin, 2012). IS has its roots in economics, but encompasses a diverse array of research areas, including business management, political science, organizational sciences, history, sociology, and more (Lundvall, 2013). In this regard, IS shares similarities with STS. However, IS tends to be more aligned with economic objectives, emphasizing goals like economic growth and international competitiveness. On the other hand, the STS tradition is deeply rooted in a critical sociological perspective, viewing science as serving broader societal objectives (Lundvall, 2013, p. 48).

The study of innovation can be traced back to the early works of Joseph Schumpeter (1943), who is commonly recognized as the father of modern innovation theory. He was a contemporary of the classical sociology of knowledge, but as an economist, he was more interested in economic development. He introduced the concept of innovation as the driving force behind the 'capitalist machine.' Schumpeter viewed entrepreneurs as agents of change, disrupting existing economic structures by introducing innovation through a process he termed 'creative destruction.' He provides basic ideas related to innovation which are central to the current innovation literature (Bontems, 2014; Fagerberg, Fosaas, et al., 2012) such as definition of innovation as new combinations of existing knowledge and resources; the distinction between invention (new ideas) and innovation (implementing these in practice); the classification of innovations into product, process and organizational innovation, and the distinction between revolutionary, radical and marginal or incremental innovations. Because he highlighted the crucial role of entrepreneurs in economic growth, his research is also acknowledged as an important source for entrepreneurship literature (Landström et al., 2012). Although both innovation and entrepreneurship share the common origins in Schumpeter's work, and are even used interchangeably (Acs & Audretsch, 2005), they have 'evolved over time as two largely separate research fields' and remain 'surprisingly disconnected' (Landström et al., 2012, pp. 117–172). The reason can be found in different directions of evolution. Innovation theory developed towards evolutionary economics and 'system on innovation' focused on institutional set up for encouraging innovation ignoring completely the

role of individual entrepreneurs who are mainly personified in big industries and large corporations.

In addition to Schumpeter, important frameworks that have influenced today's IS are growth theories which promoted intangible production factors embodied in technological change as primarily responsible for economic growth. By contrast to the neoclassical theory relying on the Nobel prize-winning exogenous growth model (Solow, 1957) in which technological change or innovation is treated as a disembodied and unexplained external part ('residual'), innovation plays a more explicit role in economic growth within the endogenous growth theory pioneered by Romer (1990). According to Romer, technological innovation is driven by endogenous factors such knowledge or research which presents inexhaustible human ideas and enables, unlike the Solow model, sustainable economic growth. His view followed a strong argument for governmental spending on basic science as a driver of technological advances already coming from the Solow model which influenced science and innovation policy for many decades including the present.

However, growth theories explore innovation as a given parameter in the context of a static and stable set of factors, often using mathematical models and missed evolutionary and non-predictable nature of innovation. The theories which take into account the changing nature of innovation which are shaped by uncertain and dynamic environment are the evolutionary (Nelson & Winter, 1982) and institutional economics (Edquist, 1997; Freeman, 1987). They emerged during the economic recession caused by oil shocks in the 1970s and 1980s and see technological innovation, primarily microelectronic revolution, as a solution to a stagnant economy (Mytelka & Smith, 2002). In contrast with growth theories, which assume that innovation is a linear process that begins with scientific research and ends with its successful commercialization, evolutionary economists establish a new innovation paradigm that sees innovation as an interactive process. This process is a result of many interactions, mutual learning and feedback loops between research, companies, markets and other innovation agents (Kline & Rosenberg, 1986).

The focus of a new innovation paradigm is on interactivity, knowledge and learning among actors, giving rise to a 'systems of innovation' approach which has led to the concept of the national innovation system (NIS) (Lundvall & Lundvall, 1992, 1995; Freeman, 1988; Nelson, 1993). The concept of NIS has gained astonishing take-up and global popularity since 1990s and remains a crucial framework for national development of contemporary countries. Systemic literature reviews have identified many other scholars like K. Pavitt, G. Dosi, Z. Griliches, C. Teece, D. J., Chesbrough, H. Von Hippel, E. Mowery, E. M. Mansfield, etc., who have made significant contribution to the development of innovation studies and policies (Fagerberg, Fosaas, et al., 2012; Fagerberg & Verspagen, 2009; Martin, 2012).

NIS is usually defined in institutional terms (Freeman, 1987, p. 7) as a setup of formal institutions (organizations like universities, government bodies, business clusters, etc.) and informal institutions which are socially rooted and make of innovation system a social system which shares common cultural norms and values (Lundvall & Lundvall, 1992). Innovation is perceived as a phenomenon endogenous to society and economy, (OECD, 1992), a complex social process which is path-dependent, locationally specific and institutionally shaped (Mytelka & Smith, 2002). National innovation capacities started to be seen as processes embedded in a broader institutional context involving

socio-cultural and political factors (Furman et al., 2002). In this context, creation of innovation requires an appropriate institutional support and deliberate policy actions specific for certain environment (region, state, etc.) to facilitate the smooth flow of knowledge between different sectors engaged in innovation processes. Therefore, policy makers have been focused on establishing innovation policy, a complex system of public measures and instruments for encouraging innovation. This policy relies on interaction between sectors, primarily between science and industry. As Lundvall (1988, p. 364) notes ‘connection between universities and industry becomes a political issue’ and still makes the essence of the innovation policies in the EU, such as the newest smart specialization strategy.

The view that policy may have a role in supporting innovation has become widespread, and the term innovation policy has become commonly used, sometimes in exchange for science, technology and industrial policy (Edler & Fagreberg, 2017), which suggests that innovation and socio-economic development are becoming synonymous.

The concept of the national innovation system is increasingly losing relevance in favor of place-based and regional innovation systems (Grillitsch & Asheim, 2018; Isaksen et al., 2022), particularly within the context of the European Union’s new industrial policy inspired by the Smart Specialization Strategy (Foray, 2014; Molica et al., 2025) which emphasizes the importance of local capacities for development. Additionally, the rise of grand societal challenges places pressure on regional innovation systems to address sustainable regional development (Bugge et al., 2021) and to incorporate the perspectives of transformative innovation policy (Diercks et al., 2019; Mazzucato, 2021; Schot & Steinmueller, 2018; Weber & Rohrer, 2012). The perspective of transformative change is further elaborated in the Discussion section.

4.5. Sociology of innovation: Diffusion theory, the Triple Helix Model, socio-technical transformations

With regard to the study of innovation, sociologists preceded economists with their studies of factors that influence the spread of innovations across societies and countries, while economists were mainly focused on innovation defined as commercialized invention (Godin, 2010; Wejnert, 2002). Sociology of innovation initially dealt with the diffusion of innovations which began with Tarde’s, 1903 book (Tarde, 1903) and his idea of spreading inventions by imitation and assimilation (Bontems, 2014). However, diffusion theory experienced a take-off only forty years later, when Ryan and Gross (1943) published their results on the spread of hybrid-corn use among Iowa farmers (Rogers, 1962; Wejnert, 2002).

Although more than 4,000 research papers on diffusion were published since that time in different research fields, mainly agriculture and healthcare, the concept was finally established and popularized by Rogers (1962), a professor of communication and journalism and ‘a true founder of ‘sociology of innovation’ (Bontems, 2014, p. 45), who analyzed the diffusion of innovation through social and organizational networks. Rogers identified the five adopter categories: innovators, early adopters, early majority, late majority, and laggards. and factors of rate of innovation adoption such as communication channels, social systems, opinion leaders and change agents. Although Rogers’s

book became a classic with the last fifth edition published in 2003, the diffusion of innovation has not come into the center of sociological research.

In addition to Tarde and Rogers, scholars (Godin, 2010) typically acknowledge two American sociologists as pioneers in the field of the sociology of innovation. One of these pioneers is William F. Ogburn (1922), who developed the idea of cultural lags, or delays in adapting to new inventions, and laid down the foundation for examining the societal impacts of technological innovation. Another noteworthy figure is Colum S. Gilfillan, who authored two seminal books in 1935 (1935a, 1935b) that explored the social factors contributing to invention. It is worth noting that it was in these works that the term ‘sociology of invention’ was first introduced.

If we look beyond the individual sociological studies related to innovation typically categorized within the field STS, we can identify two prominent sociological theories concerning the role of innovation in socio-economic development. The first is the Triple Helix Model (THM), which was developed by Leydesdorff and Etzkowitz (1998) and the second is the Multilevel Socio-Technical Transition framework proposed by F. Geels (2002, 2004).

The emergence of the THM coincided with a transition phase in the evolution of science, as documented in a book edited by American sociologists S. Cozzens et al. (1989). During this period, there was a growing demand for science to become more pragmatic and closely aligned with industry in order to justify public investment. These shifts in the organization and norms of scientific practice were best characterised by the ‘New Model of Knowledge Production,’ also known as Mode 2, as articulated by Gibbons and colleagues (1994) in 1994. Mode 2 expanded upon the traditional scientific model, referred to as Mode 1, by emphasizing academic research conducted in practical applications, fostering transdisciplinarity, and encouraging collaboration with industry. Ziman (1989, p. 23) observed that we were experiencing a radical, irreversible, and global transformation in the organization, management, and conduct of science within a single generation.

The core of the THM lies in interactions and knowledge exchange among the three sectors: firms, universities, and government, often referred to as ‘helices.’ In contrast to the national innovation system (NIS), where companies typically occupy the central role in these interactions, in the THM, the university assumes a central position as a regional or local innovation initiator and organizer (Leydesdorff & Etzkowitz, 1998, p. 197). Followers of the THM established the Triple Helix Association in 2009, launched the journal Triple Helix, offered university courses, and have been organizing regular conferences for almost 20 years.

The need for universities to be actively involved in economic development prompted Etzkowitz (2008) to formulate a thesis about the second university revolution, which broadened the paradigm of a classical research (‘Humboldtian’) university with a concept of entrepreneurial university (Dabić et al., 2016, chapters 1-4). The entrepreneurial university is characterized by a ‘third’ mission focused on commercializing science, altering traditional academic norms associated with Mertonian principles, and introducing new academic norms such as knowledge capitalization, university independence, hybridization and reflexivity (Etzkowitz, 2008, p. 41).

The concept of Triple Helix has recently been broadened by the Quadruple and Quintuple Helix Innovation Systems created by Elias Carayannis and David Campbell

(Carayannis et al., 2022, p. 2274). The Quadruple Helix model assumes that innovation results not only from the interplay of government, university, and industry policies but also from the significant roles played by civil society and media which incorporate the socio-cultural dimensions of innovation and highlight the importance of sociological study of innovation. The Quintuple Helix model further emphasizes the importance of environmental and sustainability dimensions. There is considerable debate among scientists regarding the necessity of expanding the number of helices toward the n-tuple innovation model (Leydesdorff, 2012). Some argue that adding more helices does not create new dynamics among actors, as these can always be decomposed and recombined into the fundamental Triple Helix-centered approach (Leydesdorff & Smith, 2022). Others, however, contend that the classical Triple Helix model is static and insufficient to explain sociotechnical transitions, asserting that additional helices are necessary (Park & Stek, 2022)

The theory of sociotechnical transitions is central to the Multilevel Perspective (MLP) on socio-technical transitions, introduced by Frank Geels (2002, 2004), a Dutch professor specializing in system innovation and sustainability. This framework is classified here as a sociological theory.

This classification stems from Geels' expansion of the concept of technological regimes (or trajectories) originally proposed by evolutionary economists like Nelson and Winter (1982) with sociological principles related to social rules, norms, and practices in understanding the transition from one socio-technical system to another (Geels, 2002, 2004). MLP posits that transitions occur through interactions among three distinct levels of analysis: a broad context of the landscape, a sociotechnical regime that includes social categories such as power, rules, or norms, and niches, protected spaces insulated from the influence of the dominant regime, where innovations usually occur. Innovation can challenge the existing regime and enable the transition from one socio-technical system to another. Understanding the interplay among these three levels is key to comprehending how innovation and socio-technical transitions occur.

5. Discussion

The sociology of innovation is perceived as an evolving scientific field, characterized by increasing perspectives on its own growth and evolution into an independent scholarly sub-discipline of sociology. This view is rooted in the historical evolution of the innovation concept across various social disciplines, which concurrently influenced the development of the sociology of innovation as a research field.

By examining the knowledge foundation of the sociology of innovation, four overarching theoretical frameworks were identified that make significant contributions to the field, along with three specific subfields within the narrower scope of the sociology of innovation (as shown in Table 1). In this context, the origins of the sociology of innovation can be traced back to the classical sociology of knowledge, followed by its extension into the domains of the sociology of science and sociology of scientific knowledge.

Even if it may appear that classical sociology of knowledge is outdated, esoteric, and remote from contemporary life, many of its ideas transcend the historical moment and still hold relevance today. One such example is the ongoing debate between Marx's prediction that technological innovation will impoverish workers, and

neoclassical economists' belief that innovation boosts productivity, leading to higher wages and improved living standards, which remains unresolved. Despite significant improvements in living standards in the past century, supporting the neoclassical viewpoint, debates on the connection between the 'material base and superstructure' are resurfacing spurred by accelerated neoliberalism (Burawoy, 2021) and digitalization of work (Dabić et al., 2023). Bearing in mind reproduction of the precariat through migration and digital platforms, the question of whether these material changes will create social, political, and institutional transformations leading to a fairer, more democratic society remains uncertain. Furthermore, Ragnedda (2017) showed how key distinctions identified by Weber (1922/1978/1922) about social inequality related to cultural aspects (status/prestige) and political aspects (group affiliations) are still significant for producing new forms of digital inequality or 'digital divide'. The view of Mannheim (1928/1960/1928) about the public relevance of sociology which should remedy value-neutral professionalism and partisan activism remains a critical question to the current debates about the identity of sociology as a discipline (Matić, 2017).

SSK, which emerged in the late 1970s as a counterbalance to Merton's functionalism, aimed to investigate the social construction of scientific knowledge itself. It created a platform for the institutionalization of STS, which expands the topic of social construction of science to the construction of technology and innovation. Over the past few decades, STS has evolved in an umbrella research field which integrate social fields such as Sociology of technology, Social shaping of technology (SST) (Williams & Edge, 1996, p. 892), and the Science, technology and society, which was identified with STS during its early years (Martin et al., 2012). Sociology of science and technology, dominated by Khun and Latour, was one ingredient in formation of STS and remain just one component of STS, alongside other disciplines such as business research, economics, humanities, management, and scientometrics (Law, 2008; Martin et al., 2012).

Besides the sociology of diffusion that closely aligns with classical sociology of knowledge in terms of its historical emergence, two other contemporary sociological theories attempt to delineate the field of sociology of innovation from STS. One is the Triple Helix model, which remains a valuable tool for understanding the relationships between academia, industry, and government in fostering innovation and economic development, and another one is Geel's theory of Multilevel perspective (MLP) on socio-technical transitions. MLP incorporates social categories such as norms, values and rules in understanding the transition from one socio-technical system to another. MLP is receiving increasing political and academic attention in the context of sustainable development and climate change because it provides a comprehensive framework for understanding how socio-technological transitions may unfold, considering both technological and societal factors.

Innovation studies (IS), much like STS, have evolved into a global research community since the 1960s, marked by a rich diversity of theoretical perspectives, and substantially contributes to the evolution of the concept of innovation. They emphasise the crucial role of innovation in enhancing business competitiveness and national economic growth and practically equate socio-economic development with innovation. This discourse, and the concepts of national (sectoral, regional) innovation systems and innovation policies developed within IS, remain a prominent approach used by governments worldwide to promote socioeconomic development and growth.

Evolutionary and institutional scholars, forming a core part of the IS field, perceived innovation (technical change), unlike neoclassical economists, as an interactive process implying that a strictly economic perspective alone cannot adequately explain it. Instead, they underscored the crucial role of specific institutional, social, cultural, and historical contexts in shaping the dynamics and trajectory of innovation. This perspective challenged the conventional paradigm of innovation as a linear process and a force external to society, transforming it into a complex social phenomenon which can be managed and directed by establishing efficient innovation policy and national innovation system. Innovation scholars were, therefore, much more policy-oriented than STS scholars and aimed to offer policy recommendations for setting up institutions and mechanism that will foster innovation.

However, the idea of social capability, as well as innovation itself, remained within the existing socio-technical system based on mass production and consumption which brought the world to the edge of sustainability. Prompted by the growing global challenges (UN, 2015), some scholars, such as Soete (2019), argue that IS are in an existential crisis, while Steinmueller (2013) suggests that both IS and STS may be entering a Kuhnian stage of mature ‘normal science’, which questions their capacity to identify and effectively address specific contemporary issues.

Urgent social and economic issues such as climate threats, depletion of natural resources and enormous social and ecological inequalities have motivated numerous scholars to reconsider the role of innovation and innovation policy in accordance with the transition of socio-economic systems towards sustainable development (Geels, 2020; Haddad et al., 2022; Soete, 2019). They advocate for supporting the concept of transformative innovation policy (Hekkert et al., 2020; Lundvall, 2024; Martin, 2016; Walsh et al., 2020) either from ‘bottom-up’ (Geels, 2020; Schot & Steinmueller, 2018; Weber & Rohracher, 2012) or ‘top-down’ approaches (Mazzucato, 2021). Transformational change or socio-technical transition in terms of Geels (2002, 2020) require a normative turn in the understanding of innovation (Diercks et al., 2019). It implies that innovation policy should not solely focus on economic competitiveness and growth but should also guide processes of transformative change towards desired societal objectives, a process defined by Weber and Rohracher (2012) as strategic directionality. The failure of directionality which pertains to the lack of the means to formulate specific development strategies aimed at addressing societal challenges is considered a new rationale for policy intervention within Frame 3 or transformative innovation policy. The innovation policy within Frame 1 addressed the market failures of the state supported science and technology after the Second World War, while Frame 2 was aimed at avoiding system failures in the production of innovations (Schot & Steinmueller, 2018). Sociology of innovation can play a pivotal role within Frame 3, aiming to transform both the perception of innovation and the formulation of innovation policy.

6. Perspectives

Periods of radical technological and social changes, such as these today, challenge existing socio-technical and economic structures. However, they are not limited to markets and business but also disrupt social relations, human cognition, paradigms, values and institutions. The history of social thoughts clearly pointed out that long-term economic

transformation (the ‘economic turn’) is highly interdependent with the ‘social turn’, establishing new ideologies, narratives and discourses as well as a new system of values, behavior, habits, beliefs and ethics. Social turn assumes, in addition, re-definition of social structures and institutions such as education, human and working rights, gender, migrations, family, sexuality, leisure, and more. Following this line, the answer to the question about the future of sociology of innovation is closely tied to the ongoing debate about sociology’s role in society, initiated by Burawoy’s (2005) call for a more active public-oriented sociology to restore its ruined social relevance and visibility. His predecessors asserted that sociology, since the 1980s, has been in a certain amount of crisis due to the loss of disciplinary coherence (Matić, 2017, p. 179) while Burawoy holds that current crisis of sociology does not come from a lack of theoretical integrity but from the social irrelevance of sociology and its low social visibility. He claimed that sociologists exchanged the discipline’s ‘original passion for social justice, economic equality, human rights, sustainable environment, political freedom or simply the better world’ for academic credentials, and calls for recovering its original mission to defend society (Burawoy, 2005, p. 260). Therefore, he suggests creating an organic public sociology that will restore sociology’s relevance through its direct (even partisan) civic or political engagement in order to defend social and human rights that are destroyed by the ‘market tyranny and state despotism’ (Burawoy, 2005, p. 287). Public sociology, to Burawoy, inspires the renewal of sociology as a discipline (Burawoy, 2021, p. 14).

If we leave aside his concept of ‘activist’ public sociology often criticized for politicizing sociology (Matić, 2017), Burawoy’s deliberation about sociology as a science focused on the betterment of social fabric provides a useful lead for the sociology of innovation as a science focused on social aspects of encouraging innovation to overcome global societal challenges.

Burawoy’s view that social sciences are not a melting pot of disciplines is also of a high relevance for reflections on the future of sociology and innovation. Each discipline has, to Burawoy, its specific standpoint: for economy it is market expansion, for political science is the state, while the standpoint of sociology is civil society and the defense of the social. The specific mission of sociology of innovation is therefore to examine social roots of various phenomena related to innovation from the standpoint of civil society to promote the interest and betterment of the public. Although the concept of civil society is contested and at times subject to reactionary and xenophobic ideologies, it remains the sole space for the emancipation of citizens. Consequently, it can be asserted that the research focus of the sociology of innovation is on social processes that guide society towards human-centered innovation for the benefit of civil society and aligned with the principles of social equality and moral responsibility.

An important step towards sociological research of contemporary societal challenges is made by environmental and sustainability sociology well described in the book of J. Hannigan (2006) and promoted by specialized journals like *Environmental Sociology* or *Environmental Innovation and Societal Transitions*. However, sociological research of innovation goes beyond the social studies of sustainable and environmental innovation. The primary drivers for selecting research themes within the sociology of innovation emerge from the public, which experience ‘social challenges’ prompted by innovation such as precarity, social exclusion, technological unemployment, pollution, climate change and more.

Following Burawoy's distinction between professional, policy and critical sociology (in addition to public sociology) as well as Wallerstein's (2007) distinction between intellectual, moral and political function of sociology, the main function of the sociology of innovation may include the following components: analyses of innovation production and diffusion to provide expertise and legitimacy of the theory of innovation (professional or intellectual function), evaluation of social consequences and the moral implications of innovation (critical or moral function) and analyses of the best way to realize the social turn towards innovation aligned with public interests (policy or political function).

Finally, it should be taken into account that innovation is a phenomenon with two facets: it is shaped by society, but it is also socially formidable. Innovation is a social construct (Godin, 2015, p. 2) highly dependent on the socioeconomic, cultural, and political institutions of society. On the other hand, our understanding of innovation, its general paradigm, and narratives of innovation, drive the behavior of citizens, politicians, businessmen, and other social actors.

Consequently, sociological research that builds certain theoretical framework and paradigm of innovation can significantly contribute to our common conception of innovation and can influence its strategic directionality in terms of Weber and Rohrer (2012) for the benefit of the public. Sociological research of innovation can be thus translated into useful practices and promote the idea that sociology should not only study social phenomena such as innovation in this case but to produce knowledge which induce social change. Social change related to innovation corresponds with the socio-technical transition in terms of Geels and transformation of society in terms of transformative innovation policy (Mazzucato, 2021; Weber & Rohrer, 2012). This connection promotes sociology of innovation into an important factor for social betterment.

7. Conclusion

The concept of innovation, once equated with economic growth and technological progress, which has been institutionalized and legitimized over the last 60 years, is no longer taken for granted. This shift arises from the recognition that innovation has a dark side and can be harmful to both society and nature.

Consequently, scientists argue for the necessity of abandoning the conventional paradigm associating innovation solely with economic growth and business competitiveness. Instead, the focus should shift towards steering the development of innovation in the direction of a socio-technological transition aimed at fostering a sustainable society capable of addressing grand challenges, as outlined by the UN (2015).

The sociology of innovation, through its analytical, critical, and policy functions, can play a crucial role in exploring the complex interactions between innovation and social changes. This involves breaking away from the traditional paradigm of innovation rooted in what Kuhn terms 'normal science.' The 'normal' view of innovation is likely to overlook social complexities such as inequalities, group interests, inclusion, power structures, and recognition of new problem-solving approaches detached from established socioeconomic, political, and cultural narratives and practices.

Research suggests that the sociology of innovation can evolve into an independent subfield of sociology, aiming to generate knowledge capable of inducing social change.

In doing so, it could become a significant actor in guiding society towards human-centered innovation aligned with the interests of civil society, social equality and moral responsibility. Ultimately, the objective of this research is to renew the interest of sociologists and inspire their sociological imagination to investigate innovation beyond the confines of ‘normal’ science and established paradigms. The overarching goal is to prevent failures in the overall development of society and redirect it towards a more humane course.

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